



Groundwater quality evaluation using GIS and water quality index in and around inactive mines, Southwestern parts of Cuddapah basin, Andhra Pradesh, South India

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ABSTRACT

The main purpose of this study is to evaluate the groundwater quality for human consumption by developing a water quality index for the southwest (SW) part of the Cuddapah Basin. In order to study this various physico-chemical parameters were analyzed, distribution maps were prepared using Geographical Information System (GIS). Except for fluoride and nitrate, most of the chemical parameters are within the allowable range, which is due to geogenic and anthropogenic sources, respectively. Most of the heavy metals were below the permissible limit except cadmium, which might be due to anthropogenic activities. The piper diagram shows that most samples belong to the Ca^{2+} - Mg^{2+} - HCO_3^- category. Rock dominance is the predominant hydrogeochemical factor controlling groundwater chemistry. According to WQI classification, 12% of samples are not potable due to anthropogenic activities. This study proved that the combined application of GIS and WQI is very effective in accessing groundwater quality.

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1. Introduction

Mining is one of the most important sectors to generate an economy for developing countries like India. The mining industry, such as gold, coal, barite, yellow stone, asbestos, limestone, etc., is developing rapidly and has become an incentive for the country's economic growth (Brindha et al., 2016). Active mines are generating inactive or abandoned mines, and polluting groundwater media without taking proper closer remedial actions. If a mine was inactive, the dewatering facility of the mine was ceased. In the monsoon season during heavy rainfall surface water may have entered through openings such as adits, fractures, and fissures of the underground mine; such mine water contains a significant concentration of heavy metals and total dissolved solids. This may have led to higher temperature and altered pH depending on the nature of ore body and the neutral local geochemical conditions belongs to alkaline mine water. It may also contain high pollutants higher than water quality standards and have a significant impact on human fitness and the environment. Groundwater quality problems may also be due to contamination of natural and anthropogenic activities are added to pollutants into the groundwater system (Muralidhara Reddy et al., 2019; Adimalla, 2019; Adimalla et al., 2019; USEPA, 1993). The groundwater quality increasingly corresponding to depth and also alters owing to the composition of the dissolved rocks,

physical, chemical, biological, seepage and rock sediment activities. Adverse changes in groundwater quality have occurred in recent years owing to an increased in urbanization, industrialization and agricultural activities across the globe. Owing to overuse, freshwater has become a scarce asset and water is being polluted leading to adverse health implications (Elango et al., 2012; Qian and Li, 2011; Qian et al., 2012; Li and Qian, 2018, Li et al., 2018a, 2018b; Wu et al., 2014, 2017; Zhang et al., 2018; Elango et al., 2003).

Agriculture is a significant economic sector in India, 60% of the people are dependent on it. In India more than 20 million private & government tube wells are active and is utilized for drinking, irrigation purposes (Datta and Tyagi, 1996). In India 80% of groundwater is utilized for domestic purposes while 45% of groundwater is used for irrigation. Agriculture contributes for 45% gross of the national product by majority of the people (Kumar et al., 2005; Sunitha and Sudharshan Reddy, 2019; Singh, 1983). Groundwater quality information provides us the past geological history of the rocks, changes in groundwater level, groundwater recharge and discharge movement, and aquifer storage capacity of the aquifer.

The water quality index (WQI) is defined by the combined influence of different parameters of water quality as a reflection and ranking. Water quality indexes are the most valuable ways to align common public and policy maker's knowledge on water quality to control water quality (Satyajit et al., 2020; Chimankpam et al., 2019; Prasad et al., 2019; Suvarna et al., 2018). The majority of the researchers use these indices for drinking water suitability (Adimalla and Venkatayogi,

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